

Nanotechnology based non-invasive cancer diagnostics

A research team at Zhejiang University's Polytechnic Institute in China has developed a nanosensor using indium oxide with platinum and nickel enhancements, designed



Nanoflake sensors built from indium oxide with platinum and nickel detect changes in isoprene (Credit: <https://www.azosensors.com>)

to detect early-stage lung cancer by analysing changes in breath chemistry. Known as Pt@InNiOx, these nanoflake sensors detect the lung cancer biomarker isoprene at parts-per-billion (ppb) sensitivity, achieving a detection threshold of 2ppb. Testing with a portable device on patient breath samples showed a clear isoprene concentration difference between cancer and non-cancer cases, underscoring this sensor's potential in non-invasive, early cancer diagnostics for improved patient outcomes.

AI semiconductor performance with wafer technology

A team at KIMM and Sungkyunkwan University has developed a 10.16cm heterostructure semiconductor using plasma-enhanced chemical vapor deposition (PECVD). This uses transition metal dichalcogenides (TMDc) to create low-power, silicon-alternative semiconductors ideal for AI tasks. The PECVD process enabled the fabrication of tungsten disulfide and graphene heterostructures, and two-phase molybdenum disulfide, achieving reproducible, large-scale production critical for AI, with commercial scalability for 3D integrated, heat-efficient designs.

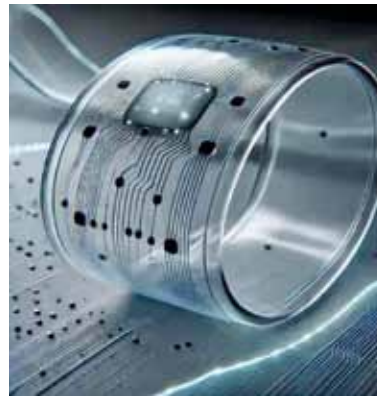
AI-powered sensor ensuring food quality

Penn State University researchers have developed an AI-driven sensor that uses a neural network and graphene-based ion-sensitive field-effect transistors to detect subtle liquid variations, advancing food safety and diagnostics.

This 'artificial tongue' analyses liquids like milk, sodas, and juices, identifying water content, spoilage, and authenticity within seconds. With a 95% accuracy rate when using machine-derived parameters, the sensor adapts to sample variations, mimicking human taste processing via the gustatory cortex. It uses Shapley additive explanations for decision transparency, enabling applications in food production, quality control, and health diagnostics, offering rapid, non-invasive assessments across industries.

Elastomer enhancing smart wearable performance

A research team at China's Ningbo Institute of Materials Technology and Engineering (NIMTE) has created an elastomer with a high di-



A high-tech, flexible wearable device made from the innovative elastomer material (Credit: <https://techxplore.com>)

electric constant and low dielectric loss, enhancing smart wearable technology. By combining ferroelectric relaxers with a rigid short chain crosslinker, they achieved a dielectric constant of ~ 35 at 1kHz and reduced dielectric loss by over 70%, now at 0.09.

This elastomer is not only highly elastic,

with 70% recovery under 60% strain, but also maintains stability under 80% strain. This material promises to advance flexible electronics, energy storage, actuation, and next-generation wearable devices.

Nanocages as anode materials for Li-ion batteries

Research from the University of Calabar investigates pristine and doped (Oxygen and Selenium) Ge₁₂C₁₂ and Si₁₂C₁₂ nanocages as potential lithium-ion battery (LiB) anodes. Using machine learning-assisted density functional theory (DFT), researchers analysed their electronic and electrochemical properties. Pristine Si₁₂C₁₂ nanocages showed high stability (2.72eV) and promising energy retention, while Se-doped Ge₁₂C₁₂ displayed superior conductivity (1.16eV). Gaussian 16 software and ML models like Lasso regression (R^2 0.99) accurately predicted cell voltages. These insights offer pathways to enhance LiB performance, particularly for electric vehicles and energy storage applications.